# Manan Chahal

mananch1 | manan-chahal | mananchahal01@gmail.com | +91 88505 76810

## Summary

Final-year Computer Engineering student with industry experience developing AI-powered applications and full-stack software. Hands-on experience in applied machine learning, computer vision, Generative AI, and RAG-based tools, alongside backend development, REST APIs, and software deployment.

## Internships

**AI/ML Intern – National Stock Exchange of India (NSE)** May 2026 – July 2026

- Worked on an automated multilingual translation system using Sarvam AI with server-side rendering, DOM traversal, and translation batching.
- Explored Redis and Elasticsearch caching with sentence hashing and cache invalidation for translated content retrieval.
- Contributed to an internal ChatGPT-style assistant using Next.js, Redis, and RAG deployed on Linux.
- Implemented speaker diarization using PyAnnote and explored ONNX/SafeTensors conversion for model deployment.

## Projects

**AI-Powered Civic Issue Reporting Platform** *Python, PyTorch, OpenCV, Node.js, Express.js, Leaflet.js*

- Fine-tuned a ResNet-34 model to verify complaint images before submission, reducing false reports.
- Built an interactive geospatial interface using Leaflet.js with distance-based dynamic loading of complaints.
- Developed backend APIs with Node.js and Express.js, implementing rate limiting to mitigate spam.

**Cloud-Based E-Learning Platform** *React, Express.js, PostgreSQL, Azure, Linux, Nginx*

- Built a full-stack web application using React and Express.js with REST APIs and auth middleware.
- Deployed on Microsoft Azure Linux VMs with SSL termination via Nginx reverse proxy.

**Glacial Flow Lines Detector (CSRE, IIT Bombay)** *Python, Random Forest, QGIS, PolSARpro*

- Project for MW Satellite Data Processing Course at IITB
- Built a Random Forest classifier to identify glacier flow lines from microwave satellite imagery.
- Processed SAR datasets using QGIS and PolSARpro to generate interpretable geospatial outputs.

## Education

| | | |
|---|---|---|
| 2023 – 2027 | B.Tech in Computer Engineering at **Vidyalankar Institute of Technology**, University of Mumbai | CGPI: **9.87** |
| 2023 | Class XII (CBSE) – Green Fingers Global School | 85.8% |
| 2021 | Class X (ICSE) – Ryan International School | 85.4% |

## Achievements

- Secured **First Place** in the subject of **Soft Computing** semester VI examinations
- **1st Place** CodeSprint (CSI-VIT) | **2nd Place** Bug Auction
- **3rd Place** Jeopardy (ITSA-VIT). | **3rd Place** Cognitive Clash College Level Hackathon
- Solved **350+** problems on LeetCode & competitive programming platforms.

## Skills

| | |
|---|---|
| **Languages** | Python, Java, JavaScript, C |
| **AI / ML** | PyTorch, Scikit-learn, OpenCV, LLMs, RAG, Speaker Diarization, ONNX |
| **Web / DB** | React, Next.js, Node.js, Express.js, Redis, PostgreSQL, MongoDB, Elasticsearch |
| **Cloud / DevOps** | Azure, Linux, Git, GitHub, Nginx, CI/CD |